

Validation of an Objective Keratoconus Detection System Implemented in a Scheimpflug Tomographer and Comparison With Other Methods

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Purpose: To validate a recently developed program for automatic and objective keratoconus detection (Keratoconus Assistant [KA]) by applying it to a new population and comparing it with other methods described in the literature.

Methods: KA uses machine learning and 25 Pentacam-derived parameters to classify eyes into subgroups, such as keratoconus, keratoconus suspect, postrefractive surgery, and normal eyes. To validate this program, it was applied to 131 eyes diagnosed separately by experienced corneal specialists from 2 different centers (Fondation Rothschild, Paris, and Antwerp University Hospital [UZA]). The agreement of the KA classification with 7 other indices from the literature was assessed using interrater reliability and confusion matrices. The agreement of the 2 clinical classifications was also assessed.

Results: For keratoconus, KA agreed in 92.6% of cases with the clinical diagnosis by UZA and in 98.0% of cases with the diagnosis by Rothschild. In keratoconus suspect and forme fruste detection, KA agreed in 65.2% (UZA) and 100% (Rothschild) of cases with the clinical assessments. This corresponds with a moderate agreement with a clinical assessment ($\kappa = 0.594$ and $\kappa = 0.563$ for Rothschild and UZA, respectively). The agreement with the other classification methods ranged from moderate ($\kappa = 0.432$; Score) to low ($\kappa = 0.158$; KISA%). Both clinical assessments agreed substantially ($\kappa = 0.759$) with each other.

Conclusions: KA is effective at detecting early keratoconus and agrees with trained clinical judgment. As keratoconus detection depends on the method used, we recommend using multiple methods side by side.

Key Words: keratoconus, detection, Pentacam

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Keratoconus is a bilateral and asymmetric noninflammatory ectatic disease in which the cornea assumes a conical shape because of progressive thinning,¹ resulting in irregular astigmatism and decreased vision. The estimated prevalence of keratoconus is 54/100,000.² The condition has a typical onset at puberty, and it may progress until after the fourth decade of life, showing progression in 88% of subjects up to 19 years old.³ Its etiology remains unknown but is associated with genetic⁴ and environmental factors, such as Down syndrome and eye rubbing.⁵ Similar forms of ectasia can occur as a complication of refractive surgery in a period of up to 10 years after surgery.⁶ Here, the most important risk factors are low central corneal thickness or residual stromal bed, high myopia, and young age.⁷ Moreover, early keratoconus detection may help to avoid progression of the condition by corneal cross-linking,⁸ thus preserving the patient's visual acuity at a relatively good level. For this reason, it is important to diagnose keratoconus in a manner that is objective, specific, and highly sensitive to the early forms of the condition, especially in refractive surgery candidates.⁹

Many authors in the literature have proposed methods to detect keratoconic corneas using combinations of parameters or specifically designed indices, such as the Score by Saad and Gatineau,¹⁰ the Belin/Ambrosio Enhanced Ectasia Display (BAD-D),¹¹ the Pentacam classification, the Cone Location and Magnitude Index (CLMI) by Mahmoud et al,^{12,13} Keratoconus index/Keratoconus Prediction Index by Maeda et al,¹⁴ and the Keratometry, Inferior-Superior and Astigmatism% index,¹⁵ inferior-superior (I-S) value,¹⁶ and Pachymetry, Asymmetry, Inferior-Superior Index¹⁷ by Rabinowitz. Some of these indices were compared previously by Mahmoud et al.¹⁸

In a recent work, our group developed a highly accurate artificial intelligence system called Keratoconus Assistant

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(KA),¹⁹ specifically designed to detect early corneal ectasia. This program may be installed alongside Pentacam software to perform a live assessment of measured data, and it gives a pop-up message with the probability of belonging to one or more of the given classification groups.

Because validation is essential to assess the clinical usability of KA, it is the aim of this study to verify the program by applying it to a cohort that was not involved in its development. We also compared the output of the KA with the above-mentioned methods, which are the most widely known indices for keratoconus detection.

MATERIALS AND METHODS

Patients

This work uses 131 eyes of 102 subjects measured during routine clinical practice at the Rothschild Foundation in the period between October 2015 and January 2016. The demographic details of the groups can be found in Table 1. These eyes were not used in the previous analysis that defined KA.¹⁹ Exclusion criteria were a history of ocular surgery (apart from laser correction in the refractive surgery group), corneal cross-linking, and very advanced keratoconus with corneal scarring. Rigid contact lens wearers were asked to remove their lenses at least 1 week before testing. Retrospective analysis of these data for the purpose of this study was approved by the Ethical Committee of the Fondation Ophtalmologique Adolphe de Rothschild; informed consent was not required.

Methods

All patients had a complete ophthalmic evaluation that consisted of manifest refraction, slit-lamp examination, and corneal tomography evaluation by both Pentacam HR (v1.20 r53; Oculus GmbH, Wetzlar, Germany) and Orbscan IIz (Technolas Perfect Vision, Munich, Germany).

Subjects were clinically diagnosed at the Rothschild Foundation based on slit-lamp examination and Orbscan

topography and classified into one of the following groups: normal (N), keratoconus (KC), forme fruste and suspect keratoconus (SS), and postrefractive (PR) surgery for myopia. Later, the same subjects were diagnosed independently by the Antwerp University Hospital (UZA) based on the subjects' Pentacam tomography and classified into the same groups as before. Each clinical diagnosis was performed according to the standards and work flow of the center that performed it. As such, there was a major difference between both centers that must be kept in mind, in that the diagnoses were based on 2 different tomographers (Placido and Orbscan in Rothschild, and Pentacam at UZA). Because these devices are known to differ in their measurements,²⁰ this may lead to a difference in interpretation in early keratoconus. In both centers, forme fruste was defined as the tomographically normal companion eyes of eyes with obvious keratoconus, and keratoconus suspect eyes were the dubious cases in which minor signs of asymmetry were present, but too little to consider it as early keratoconus. The tomography data were also used to classify the corneas using the following methods:

Keratoconus Assistant

KA is an open-source program that can be installed alongside Pentacam software.¹⁹ It detects new measurement or the opening of an old examination by monitoring and reading the comma separated value files in the Pentacam root folder that are updated each time results are displayed on the screen. From these files, 25 topography and tomography parameters of the anterior and posterior corneal surfaces (see Supplemental Digital Content 1, <http://links.lww.com/ICO/A515>) are extracted and fed into a support vector machine algorithm that automatically classifies corneal patterns and notifies the user by a pop-up showing the probability that the cornea has keratoconus, forme fruste or suspect keratoconus, PR surgery, normal, or regular astigmatism. For this study, normal eyes and eyes with regular astigmatism were considered as one group because none of the other classification methods studied here included a regular astigmatism class.

TABLE 1. Description of the Study Groups According to the UZA and Rothschild Classifications

Group (No. Subjects)	Age, yr	K _{mean} , D	Astig, D	K _{max} , D	Pachy _{min} , μm
UZA classification					
KC (54)	34.0 ± 10.5 (16–60)	45.4 ± 2.5 (40.6–51.2)	2.4 ± 1.2 (1.1–5.5)	49.1 ± 4.2 (41.5–59.9)	481.9 ± 38.9 (398–562)
N (44)	33.4 ± 13.1 (18–75)	44.4 ± 1.3 (42.1–47.4)	1.0 ± 1.0 (0.5–5.5)	45.3 ± 1.5 (42.4–49.0)	554.6 ± 26.9 (488–597)
PR (10)	38.2 ± 12.7 (22–67)	38.6 ± 1.6 (36.3–41.0)	0.2 ± 0.1 (0.0–0.4)	43.7 ± 1.7 (41.3–46.5)	485.2 ± 32.8 (422–429)
SS (23)	39.0 ± 17.2 (21–78)	44.3 ± 1.4 (42.4–47.7)	1.7 ± 1.2 (0.4–5.1)	45.5 ± 2.2 (43.1–51.0)	531.7 ± 39.7 (435–591)
Rothschild classification					
KC (51)	33.5 ± 10.9 (16–60)	45.0 ± 2.3 (40.1–50.5)	2.5 ± 1.2 (1.1–5.5)	49.7 ± 3.9 (44.4–59.9)	477 ± 37 (398–562)
N (61)	36.0 ± 15.5 (18–78)	45.2 ± 1.4 (42.4–49.0)	1.0 ± 0.9 (0.4–5.1)	44.0 ± 1.2 (42.0–47.0)	551 ± 28.8 (483–597)
PR (11)	38.4 ± 12.1 (22–67)	38.7 ± 2.0 (35.7–42.2)	0.3 ± 0.3 (0.0–1.2)	43.6 ± 1.7 (41.3–46.5)	498.3 ± 33.9 (429–530)
SS (8)	32.3 ± 8.3 (22–40)	42.5 ± 1.3 (40.5–45.0)	1.4 ± 0.9 (0.4–3.2)	43.5 ± 1.3 (41.5–45.9)	512.5 ± 29.8 (484–553)

Mean ± SD (range).
Astig, anterior corneal astigmatism; KC, keratoconus; K_{max}, maximum keratometry value; K_{mean}, mean keratometry; N, normal; Pachy_{min}, pachymetry at thinnest point; SS, keratoconus suspect.

Score

This is an artificial intelligence system that uses a linear combination of 12 anterior and posterior corneal parameters obtained with the Orbscan II to derive a compound score.^{10,21} This score is used as a discriminator by associating a positive score (>0) with a suspicious or keratoconic cornea and a negative score (<0) with a normal cornea. The Score value was retrieved directly from the Orbscan.

BAD-D

The BAD-D is a screening tool incorporated in Pentacam software.¹¹ It is based on the analysis of pachymetry progression, as well as anterior and posterior corneal elevation maps. Its output is a total deviation value, or a global “D,” which is calculated using 5 specifically defined parameters and a regression analysis and compared with a large normative database. The cutoffs for the total D are as follows: $D < 1.6$ for normal, D between 1.6 and 3 for suspicious, and $D > 3$ for keratoconus. The D value was retrieved directly from the Pentacam.

Pentacam Classification

This is an adaptation of the Amsler–Muckenhirn²² standards that classifies corneas based on anterior topography into normal [minimum radius of curvature ($R_{\min}$) greater than 7.8 mm], keratoconus suspect ($R_{\min}$ between 7.8 and 6.7 mm), keratoconus (a scale from 1 to 4, in which level 1 has $R_{\min}$ values between 7.5 and 6.5 mm, level 2 between 6.9 and 5.3 mm, level 3 between 6.6 and 4.8 mm, and level 4 less than 5 mm), corneal surgery, and abnormal. This classification is available on screen and in Pentacam export files.

Cone Location and Magnitude Index

The CLMI detects keratoconic patterns using anterior corneal topography maps.¹² The algorithm searches within the mapped area to find the steepest-area local region, which is then compared with the rest of the maps to determine whether or not this local maximum region is consistent with the corneal asymmetry associated with keratoconus. From CLMI, a discriminating parameter, percent probability of keratoconus, can be derived, with percent probability of keratoconus greater than 0.45 corresponding to keratoconus, between 0.20 and 0.45 to keratoconus suspect, and below 0.20 to normal eyes. Although the algorithm was later updated to include the posterior surface and pachymetry maps as well,¹³ we were unable to implement this version because of information missing from the article.

I-S Value

The I-S value measures inferior versus superior corneal dioptric asymmetry.¹² An I-S value below 1.4 D is normal, between 1.4 and 1.9 D is suspicious, and > 1.9 D suggests keratoconus. This value is provided in the Pentacam export files.

KISA% Index

This index uses a combination of keratometry, I-S value, astigmatism, and skewed radial axis to distinguish between regular and irregular astigmatism.¹⁵ A KISA% value

lower than 60% is normal, in the range of 60% to 100% suggests keratoconus, and a value exceeding 100% strongly suggests advanced keratoconus. This parameter was calculated based on the Pentacam export files.

Pachymetry, Asymmetry, Inferior-Superior Index (PA/I-S)

It is defined as the minimum pachymetry value divided by 4 times the I-S value, which corresponds with keratoconus if it is below 100 and with a normal eye otherwise.¹⁷ This value was calculated using the 2 variables from the export files of the Pentacam.

Statistics

In a first step, the ability of each method to accurately classify corneas compared with the clinical classifications was assessed using confusion matrices. The agreement between KA and the other methods was assessed using Cohen kappa interrater reliability.²³ This quantifies the level of agreement among the classifications by various methods and corrects for the level of agreement that would be expected by chance to form a standardized index of the interrater reliability that may be generalized across studies. The degree of agreement is determined by cross-tabulating ratings for pairs of methods, and the agreement expected by chance is determined by the marginal frequencies of each rater's ratings. Kappa is then calculated as $\kappa = P(a) - P(e) / 1 - P(e)$, where $P(a)$ denotes the observed percentage of agreement, and $P(e)$ denotes the probability of expected agreement due to chance. Kappa ranges between -1 and $+1$, with 1 indicating perfect agreement, 0 indicating complete randomness, and -1 indicating “perfect” disagreement.²⁴ Analyses were performed with SPSS (v. 23.0 IBM; Armonk, NY) on a one-on-one basis, with the KA against the others. In case a method did not contain the same groups as the KA, the subjects from those groups were excluded from the analysis. The number of subjects may therefore vary between analyses.

In addition, the accuracy of the KA using the new data set was determined using receiver operating characteristic (ROC) curves. Typically, the test accuracy in an ROC curve is reported in the form of the area under the curve (AUC), which is classified as excellent ($AUC = 0.9-1.0$), good ($AUC = 0.8-0.9$), fair ($AUC = 0.7-0.8$), poor ($AUC = 0.6-0.7$), or fail ($AUC = 0.5-0.6$).

RESULTS

Agreement Between Clinical Assessment and Other Methods

The confusion matrices of the 2 clinical assessments (UZA and Rothschild) and the different classification methods are shown in Table 2. It is seen that UZA and Rothschild classified quite a few eyes differently, especially keratoconus suspect and normal eyes: for UZA, there were 23 SS and 44 N, whereas for Rothschild, there were 8 and 61, respectively. This highlights the differences in equipment and criteria for forme fruste keratoconus used in the 2 clinical assessments.

TABLE 2. Confusion Matrices (Actual vs. Predicted Class) of the Class Given by the 2 Subjective Classifications and the Other Methods: KA, Pentacam, KISA% Index, BAD-D, I-S Value, CLMI, Score, and PA/I-S

Classification Method	Actual Class							
	UZA Classification				Rothschild Classification			
	KC (n = 54)	SS* (n = 23)	N (n = 44)	PR (n = 10)†	KC (n = 51)	SS* (n = 8)	N (n = 61)	PR (n = 11)†
KA								
KC	50	1	2		50		3	
SS	4	14	23		1	8	31	1
N		8	19	2			27	2
PR				8				8
Pentacam								
KC	32				32			
SS	8				5	3		
N	13	23	44		13	5	61	1
PR	1			8	1			8
ABN				2				2
BAD-D								
KC	42				42			
SS	10	6	4		9	2	9	
N	2	17	40			6	52	
I-S								
KC	31				31			
SS	3				3			
N	20	23	44		17	8	61	
CLMI								
KC	53	15	18		50	7	28	
SS	1	5	15		1	1	19	
N		3	11				14	
KISA%								
KC	17	2	14		15	2	16	
SS	1	3	7		2	1	8	
N	1	14	22		1		35	
Error	35	4	1		33	5	2	
Score								
KC	50	6	5		49	3	9	
N	4	17	39		2	5	52	
PA/I-S								
ABN	40		1		39	1	1	
N	14	23	43		12	7	60	

*Includes forme fruste keratoconus as the nonsymptomatic companion eye of an eye with clear keratoconus.

†Only available for the KA and Pentacam classifications.

ABN, abnormal; KC, keratoconus; N, normal; PA/I-S, Pachymetry, Asymmetry, Inferior-Superior Index; PR, postrefractive surgery with myopic correction; SS, keratoconus suspect and forme fruste keratoconus.

Compared with the clinical UZA classification, KA classified 8/23 SS as N, and 23/44 N as SS. KA is, therefore, on the conservative side (ie, a high number of false positives). The Pentacam classification performs well for N eyes (100% classified correctly), but also classified a number of keratoconic eyes as normal (13/54 KC and 23/23 SS), which suggests that it is relatively insensitive to subtle signs of early keratoconus. This could be due to the fact that the Pentacam classification only takes the anterior corneal surface into account. BAD-D shows the same tendency as the Pentacam classification and classifies 17/23 SS eyes as N and 10/54 KC as SS. I-S performs similarly, classifying 20/54 KC and 23/23 SS as N. This

suggests that these tools are rather insensitive to early signs of corneal ectasia. We were unable to calculate KISA% in 31.9% of the cases. Among the eyes that could be calculated, quite a few normal eyes were classified as KC (14/44), as well as 14/23 SS and 5/10 post-LASIK eyes. For CLMI, 15/23 SS eyes were classified as KC, 15/44 of the N eyes as SS, and 18/44 N eyes as KC. Score agreed with the clinical UZA classification in 50/54 of the KC eyes, but 17 of the 23 SS eyes were classified as normal. Finally, PA/I-S classified 40/54 of the KC eyes as abnormal, while classifying all suspect eyes as normal (23/23). Similar results were seen comparing the automatic indices with the clinical Rothschild classifications (Table 2).

Agreement Between KA and Other Methods

The agreement between the KA and all the other methods is given in Table 3, expressed as Cohen kappa (κ), standard error, and kappa confidence interval. A moderate agreement is found between KA and the Rothschild clinical assessment ($\kappa = 0.594$), as well as KA and the UZA clinical assessment ($\kappa = 0.563$). The agreement between KA and Score is also moderate ($\kappa = 0.432$). The agreement was lowest with KISA% ($\kappa = 0.138$), a value that almost corresponds with pure chance.

Finally, we performed an ROC curve analysis, comparing the accuracy of the KA classifications for each group with respect to the clinical UZA classification. The AUC value was 0.9978 for discriminating KC versus N, and 0.9355 for FF versus N, which is rather high compared with previous studies in the literature (Table 4).

DISCUSSION

KA had a high accuracy and sensitivity in discriminating KC from N eyes (98.9% and 99.1%, respectively) at the time of its development and first evaluation.¹⁸ When discriminating forme fruste keratoconus (ie, a tomographically normal eye with manifest KC in the fellow eye) from normal eyes, the accuracy went down to 93.1%. Finally, when 5 groups were included in the analysis (keratoconus, forme fruste keratoconus, regular astigmatism, PR surgery, and normal), which is the situation closer to clinical practice, the global accuracy was 88.8%. This study aimed to validate these results by applying KA to a new population that was not used during its development and agreement with other known detection methods.

KA was found to agree reasonably well with the Rothschild and UZA clinical classifications (interrater agreement $\kappa = 0.594$ and $\kappa = 0.563$, respectively), which shows that the automated screening tool agrees well with the judgment of trained clinical professionals. It did, however, provide a large number of false positives, classifying 23/44 of UZA and 31/61 of Rothschild normal subjects as SS. This

TABLE 3. Interrater Agreement Between KA and Other Methods

Classification Method	Categories	Total No. Eyes per Analysis	κ	SE CI (min, max)
Subjective Rothschild	KC, SS, N, PR	131	0.594	0.048 (0.500–0.688)
Subjective UZA	KC, SS, N, PR	131	0.563	0.054 (0.457–0.669)
Score	KC, N	136	0.432	0.045 (0.344–0.520)
BAD-D	KC, SS, N	136	0.374	0.055 (0.266–0.482)
Pentacam	KC, SS, N, PR	131	0.354	0.048 (0.260–0.488)
PA/I-S	ABN, N	136	0.324	0.044 (0.238–0.410)
I-S value	KC, SS, N	136	0.279	0.042 (0.197–0.361)
CLMI	KC, SS, N	136	0.219	0.051 (0.119–0.319)
KISA%	KC, SS, N	91	0.138	0.068 (0.005–0.271)

ABN, abnormal; CI, confidence interval; PA/I-S, Pachymetry, Asymmetry, Inferior-Superior Index; SE, standard error; κ , Cohen kappa.

may be due to the relatively low number of suspect eyes KA was trained with and the close similarity of this group to normal eyes. Another possible reason could be misclassification of suspect eyes by clinical experts who consider subtle

TABLE 4. Summary Studies that Discriminate Between Keratoconus (KC), SKC, Suspect Keratoconus (KCSS), Forme Fruste Keratoconus (FF), and Normal (N) Eyes

Study			Variable/Index	AUC
Author	Year	Groups	Name	
Twa	2005	KC vs. N	Decision tree	0.97
			CLMI	0.92
			KISA%	0.92
			Modified Rabinowitz-McDonnell index	0.91
Saad	2010	KC vs. N	Discriminant function	0.99
Ucakhan	2011	KC vs. N	Anterior elevation depression differences from best fit sphere	0.947
			Minimum radius of curvature	0.943
			Average pachymetry progression index	0.943
			Discriminant function of Zernike corneal and ocular coefficients	0.985
Qin	2013	KC vs. N	Logistic regression formula	0.975
			Keratoconus risk score	0.949
Rabinowitz	2014	KC vs. N	Inferior-superior dioptric asymmetry	0.99
			Coma	0.96
			Minimum pachymetry	0.95
			Third-order RMS	0.83
Reddy	2014	Early KC vs. N	Total RMS	0.82
			Minimum pachymetry	0.79
Reza	2015	KC vs. N	Anterior elevation	0.970
			Posterior elevation	0.977
Hashemi	2016	KC vs. N	Belin-Ambrosio display	0.99
			Mean keratometry	0.86
Current	2017	KC vs. N	KA	0.9978
De Sanctis	2008	SKC vs. N	Posterior elevation (standard)	0.80
			Posterior elevation (modified)	0.72
Saad	2011	FF vs. N	Discriminant function	0.98
Ucakhan	2011	SKC vs. N	Minimum pachymetry progression index	0.820
			Average pachymetry progression index	0.842
			Maximum pachymetry progression index	0.840
			Tilt	0.998
Gi Hyun	2014	FF vs. N	Superotemporal-inferonasal asymmetry at 4 mm	0.809
			Anterior elevation	0.698
Reza	2015	SKC vs. N	Posterior elevation	0.698
			Belin-Ambrosio display	0.86
Hashemi	2016	KCSS vs. N	Index of vertical asymmetry	0.86
			Index of surface variance	0.80
			KA	0.9355

SKC, subclinical keratoconus; AUC, area under the curve; RMS, root-mean-square.

keratoconic signs as normal or the other way around. Although these false-positive cases are less clinically dangerous than false negatives, a high rate of false positives may cause physicians to lose interest in the warnings of the KA, prompting us to continue working to reduce the amount of false positives in the next versions of the program.

Score¹⁰ is an index developed to detect very early forms of keratoconus and uses a cutoff value to distinguish between normal and suspicious or keratoconic corneas. During the validation of this method in independent data, Saad and Gatinel²¹ found an accuracy of 91.7% detecting corneas at risk for developing post-LASIK ectasia, which is in disagreement with our findings, in which only 6/23 of the suspects were detected according to the UZA clinical assessment and 3/8 according to the Rothschild assessment.

PA/I-S was recently proposed by Rabinowitz et al¹⁷ as a sensitive parameter for differentiating between keratoconus suspects or forme fruste keratoconus and normal corneas. They were able to classify correctly all 7 of their 7 forme fruste eyes and 15 of their 16 suspect eyes. This is in contradiction with our findings, in which all suspect eyes were classified as normal by PA/I-S with regard to the UZA and 7/8 to the Rothschild assessment. The reason for this discrepancy could be the definition of the SS group, as Rabinowitz described FF as corneas with skewed radial axes and SS as presenting mild inferior steepening, which is more lenient than the definitions used in this work.

Meanwhile, the other early keratoconus screening tools, such as Pentacam and I-S value,¹⁶ classified all suspect eyes as normal. For the BAD-D,^{11,25} this was slightly better, with 17/23 suspects classified as normal. To a lesser extent, this was also seen for manifest keratoconus, with 13/54, 20/54, and 14/54 of the keratoconic eyes classified as normal for the Pentacam, I-S value, and PA/I-S, respectively. As in a clinical setting, such misclassifications may be dangerous and could potentially lead to complications (eg, postrefractive ectasia); caution is warranted when using these indices for prerefractive screening.

Note that, although both clinical methods substantially agreed with each other ($\kappa = 0.759$), there were still considerable discrepancies between them because of differences in equipment and definition of forme fruste keratoconus. Considering that physicians around the world partially depend on these indices to identify patients at risk for postrefractive ectasia, it is important that a clear definition of forme fruste keratoconus becomes available in the literature. Furthermore, the examples above show that the treatment patients will undergo may depend on their ophthalmologist's choice of topographer, detection index, and cutoff values. Therefore, it is very important in clinical practice to stick to the same device and criteria when screening patients' previous refractive surgery.

Another important aspect regarding these indices is choosing the right cutoff values to distinguish between normal eyes and (early) keratoconus. For example, in a study by Villavicencio et al,²⁶ the BAD-D was tested in an independent population of normal and keratoconic eyes. They found a false-negative rate of 1% and false-positive rate of 5% when D was set to 1.65 with a 95% confidence interval, whereas setting the cutoff for D to 2.69 led to a false-positive

rate of 0%. This study did not report the performance of D in cases with early or no signs of keratoconus, however. Similarly, Rabinowitz set the cutoff value for I-S to 1.2 D¹⁶ to discriminate between keratoconus and normal eyes in 98% of the subjects, with a 5% false-positive rate. This is in disagreement with the global accuracy found in this study for I-S of 61.9%. In another work by the same author, it was stated that KISA% was the most useful computerized tool at that moment to distinguish keratoconus from normal eyes, with an accuracy of 99.6% if the cutoff was set to 100.¹⁵ The accuracy of this index classifying keratoconus suspect eyes was 75% for cutoff between 60 and 100, and early keratoconus eyes were classified as KC with a cutoff of 100 in 91.7% of the cases.¹⁵ These results disagree with the present findings, in which an accuracy of 62.9% was found for KISA% to distinguish between KC and N. This disagreement could again be due to the fact that their suspect eyes had a well-defined area of corneal steepening, in contrast to the criteria used in this study.

After evaluating the KA with regard to the clinical classification and comparing it with other widely recognized methods, we can conclude that it is an effective tool to identify eyes with forme fruste keratoconus at risk for developing corneal ectasia after refractive surgery. Moreover, it was shown to agree with trained clinical judgment. However, the false-positive rate should be reduced in the next versions of the program by increasing the sample size of the training set, especially for the suspect group.

Finally, like any other tool, KA should not be considered as a stand-alone diagnostic test but as an additional tool alongside other clinical data, such as family history, refraction, slit-lamp examination, and evolution of the corneal shape.

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